Introduction

- **Probability theory** and **Statistics** are branches of mathematics developed to deal with *uncertainty*.
- Uncertainty in computer environment
  - Arrival of jobs
  - Execution time
  - Memory requirement
  - Failure of components
  - Exposure to viruses
  - Errors in codes
  - etc

What we'll learn

- Uncertainty; probabilities of events
- Random variables
- Stochastic processes

- Application:
  - queuing systems
  - Statistical inference
  - Estimation
  - Testing hypothesis

Some definitions

- **Experiment** : An experiment is a procedure or process that produces outcomes. It is a well-defined action or series of actions that leads to observable results.
  - For example, rolling a die or flipping a coin.
- **Outcome :** An outcome is a single possible result of an experiment.
  - For example, if you roll a die, the outcome could be any one of the numbers 1 through 6.
- **Sample space**: ~ is the set of all possible outcomes of an experiment.
  - For example the sample space of rolling a die is {1, 2, 3, 4, 5, 6}.
  - Sample space is defined by S.

Some definitions

- **Event:** An event is a subset of the sample space, containing one or more outcomes.

  - For example, when rolling a die, the event of getting an even number can be represented as {2, 4, 6}.
  - **certain event** (S) = happens with certitude at each repetition of an experiment
  - **impossible event** (ϕ) = never takes place in a random experiment

Some definitions

- **Discrete Sample Space**: A discrete sample space consists of a finite or countably infinite set of distinct outcomes. Examples include flipping a coin (sample space {Heads, Tails}) or rolling a die.

- **Continuous Sample Space**: A continuous sample space consists of an infinite number of outcomes that form a continuum. These outcomes cannot be listed individually but are instead described by intervals. An example is measuring the exact height of a person, where the outcomes could be any real number within a certain range.

Some definitions

- **Example:**
  - Sample spaces with a large or infinite number of sample points are best described by a **statement** or **rule method**.

$$S = \{\, x \mid x \text{ is a city with a population over 1 million} \}$$

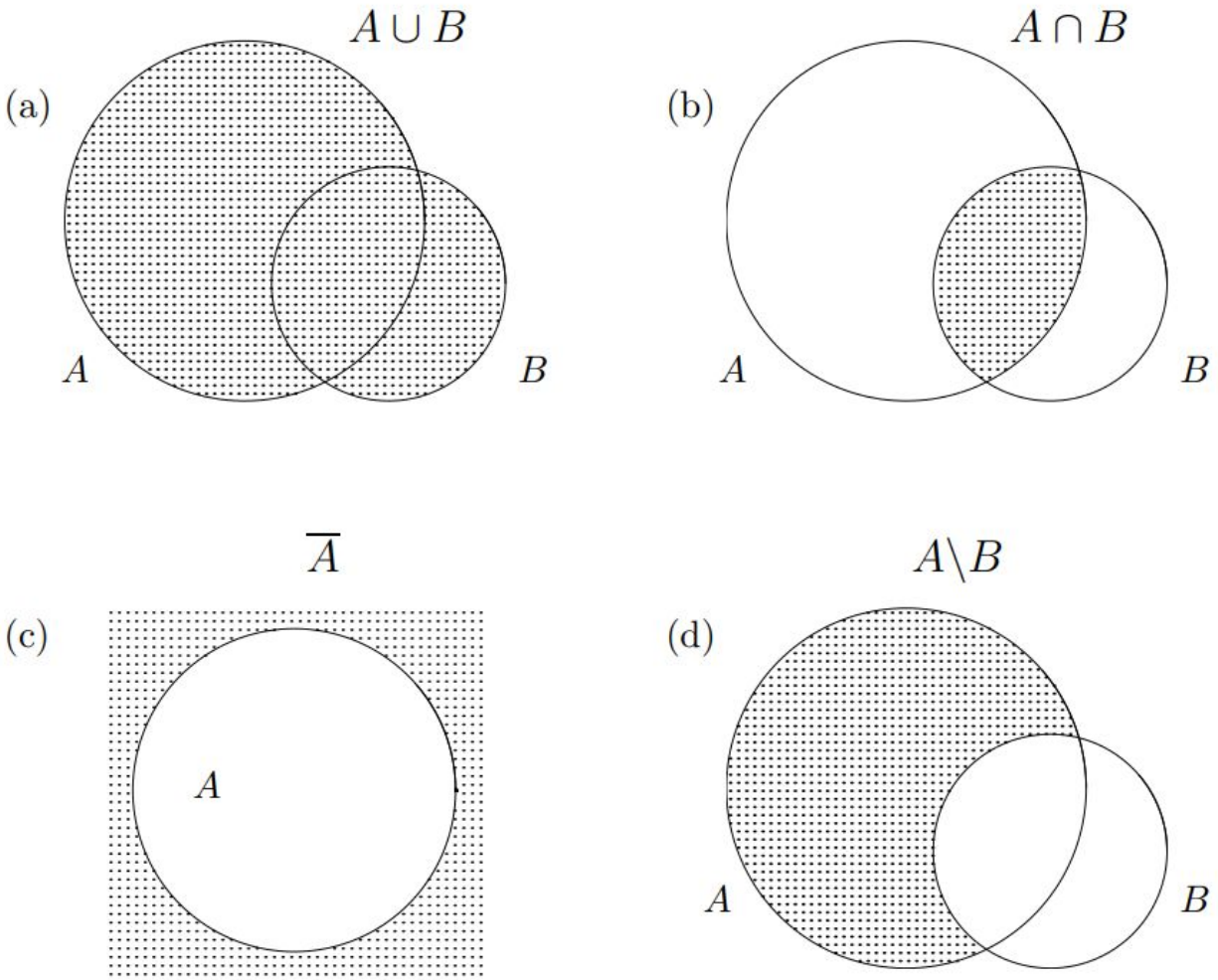
  - *S* is the set of all points (*x*, *y*) on the boundary or the interior of a circle of radius 2 with center at the origin

$$S = \{(x, y) \mid x^2 + y^2 \leq 4\}.$$

Operations with Events

- The language and concepts of **set theory** provide a natural context for the development of probability theory

Set theory	Probability theory
Sets $A, B, \ldots$	Events $A, B, \ldots$
A	Event A occurs
$\overline{A}$	Event A does not occur
$A \cup B$	At least one of A and B occurs
AB	Both A and B occur
$A \subset B$	A is a subevent of B (i.e. the occurrence of A necessarily implies the occurrence of B)


(a)
$A \cup B$
A
B
(b)
$A \cap B$
A
B
(c)
$\overline{A}$
A
(d)
$A \backslash B$
A
B

Operations with Events

- ***A \ B* (difference) =** $A \cap \overline{B}$ = the event *A* occurs but *B* does not
- Events *A, B, C,..* are **disjoint (or mutually exclusive)**
  - if ***A inters B inters C*** … = ϕ
  - disjoint events cannot occur at the same time
- Events *A, B, C, …* are **exhaustive** if **A** ∪ **B** ∪ **C** ∪ **…** = S

Commutative law $E \cup F = F \cup E$ $EF = FE$

Associative law $(E \cup F) \cup G = E \cup (F \cup G)$ $(EF)G = E(FG)$

Distributive law $(E \cup F)G = EG \cup FG$ $EF \cup G = (E \cup G)(F \cup G)$

Counting Techniques

- **Multiplication rule.** If an experiment consists of *k* components (sub-experiments) for which the number of possible outcomes are $n_1, n_2, \ldots, n_k$ then the total number of experimental outcomes (the size of the sample space) is equal to $\boldsymbol{n_1 \times n_2 \times \ldots \times n_k}$.

- The number of **permutations** of *n* distinct objects taken *r* at a time is

$$ {}_nP_r = \frac{n!}{(n-r)!}. $$

- The number of **combinations** of *n* distinct objects taken *r* at a time is

$$ \binom{n}{r} = \frac{n!}{r!(n-r)!}. $$

  - Many other permutation/combination rules

Counting Examples

- The design for a Website is to consist of four colors, three fonts, and three positions for an image. From the multiplication rule, $4 \times 3 \times 3 = 36$ different designs are possible.

- An electronic controlling mechanism requires 5 distinct, but interchangeable, memory chips. In how many ways can this mechanism be assembled by placing chips in the odd numbered positions within the controller?
  - Ans: ${}^5P_3$

- A young boy asks his mother to get 5 Game-Boy™ cartridges from his collection of 10 arcade and 5 sports games. How many ways are there that his mother can get 3 arcade and 2 sports games?
  - Ans: ${}^{10}C_3 \times {}^5C_2 = 1200$

Axioms of Probability

- **Axiom 1:** for each event A in S, $P(A) \geq 0$ (nonnegative).
- **Axiom 2:** $P(S) = 1$ (normed).
- **Axiom 3:** for a collection of mutually exclusive events $A_1, A_2, \ldots$ in S

$$P(A_1 \cup A_2 \cup \ldots) = P\left(\sum_j A_j\right) = \sum_j P(A_j)$$ (additive).

- **Probability**. A *probability measure*, or simply a *probability*, on a sample space S is a specification of numbers P*(A)* for all events A that satisfy Axioms 1, 2, and 3.
- function of an event = P(E)
  - **Domain**: events
  - **Range**: [0, 1]

Assignment of Probability

- Axioms of probability define the properties of a probability measure
  - However, they do not guide us in assigning probabilities to various events
- For problems in applied sciences, a natural way to assign the probability of an event is through the observation of *relative frequency*.
  - If an experiment can result in any one of **N** different equally likely outcomes, and if exactly ***n*** of these outcomes correspond to event A, then the probability of event A is

$$P(A) = \frac{n}{N}.$$

  - Two events *A* and *B* are **equally likely** if there is no reason to suppose that the occurrence of one is favored with respect to the other.

Assignment of Probability

- **subjective approach** to probability assignment is *relative likelihood*.
  - it is not feasible or is impossible to perform an experiment a large number of times
  - the probability assigned as a result of subjective judgement.
  - The statement ‘there is a 40% probability of rain tomorrow’
  - where the number 0.4 is assigned on the basis of available information and professional judgement

Probability Example

- A coin is tossed twice. What is the probability that at least 1 head occurs?

  $S = \{HH, HT, T\,H, T\,T\}$

  Let, coin is balanced, each outcomes are equally likely.

  we assign a probability of ω to each sample point.
  Then $4\omega = 1$, or $\omega = 1/4$.

  If A represents the event of at least 1 head occurring, then
  $A = \{HH, HT, T\,H\}$ and $P(A) = \frac{1}{4} + \frac{1}{4} + \frac{1}{4} = 3/4$

Probability Example

- A die is loaded in such a way that an even number is twice as likely to occur as an odd number. If E is the event that a number less than 4 occurs on a single toss of the die, find P(E).
- let A be the event that an even number turns up and let B be the event that a number divisible by 3 occurs.
  - Find $P(A \cup B)$ and $P(A \cap B)$.

$A = \{2, 4, 6\}$ and $B = \{3, 6\}$
$A \cup B = \{2, 3, 4, 6\}$ and $A \cap B = \{6\}$.

$P(A \cup B) = 2/9 + 1/9 + 2/9 + 2/9 = 7/9$

$P(A \cap B) = 2/9$

Probability Example

- During some construction, a network blackout occurs on Monday with probability 0.7 and on Tuesday with probability 0.5. What is the probability that blackout will appear on Monday or Tuesday?

- Then, does it appear with probability 0.7+ 0.5 = 1.2?
  - Obviously not, because probability be between [0 – 1]
- Probabilities are not additive here because blackouts on Monday and Tuesday are not mutually exclusive. P(A and B)=0. Do not occur same time. Counts twice(overestimation)
  - In other words, it is not impossible to see blackouts on both days.

Rules of Probability

- $P(\phi) = 0$
- **Addition rule**
  - If A and B are any events in S, then
    $$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$
  - For mutually exclusive events, $P(A \cup B) = P(A) + P(B)$

- **Complement rule**

  $P(A') = 1 - P(A)$.

Rules of Probability - Independent events

- Events A and B are independent if they occur independently of each other,
  - i.e., occurrence of one event does not affect the probability of other.
- Decide which pair of events are dependent or independent.

  1. Toss a coin. Then roll a number cube (die).
  2. Choose a bracelet and put it on. Choose another bracelet.
  3. Select a card. Do not replace it. Then select another card.
  4. Select a card. Replace it. Select another card.
  5. Pick one flower from a garden, then pick another.

  1,4 – independent

- For independent events – **multiplicative rule**
  - $P(A \cap B) = P(A)\ P(B)$
  - $P(E_1 \cap \ldots \cap E_n) = P(E_1) x \ldots x\ P(E_n)$

Example

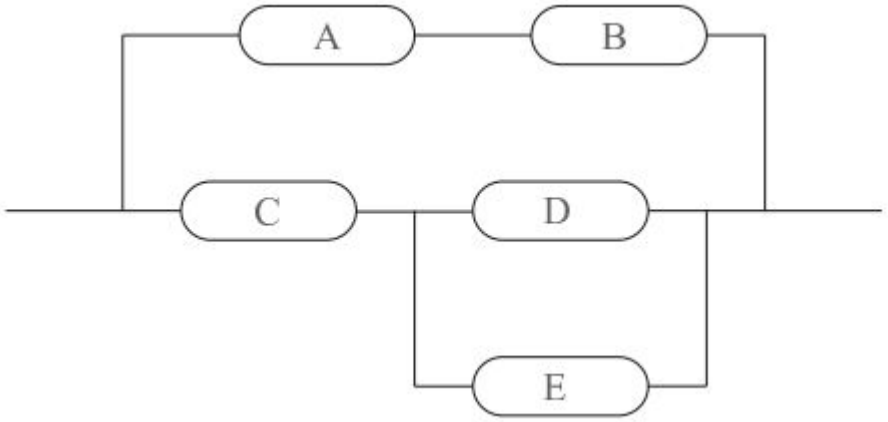
- There is a 5% probability for a hard drive to crash. Therefore, it has a backup having a 10% probability to crash, and all components are independent of each other. The stored information is lost only when all devices crash. What is the probability that the information is saved?

$$H = \{ \text{hard drive crashes} \}$$
$$B = \{\text{backup crashes} \}$$
$$P(H) = 0.05, \text{ and } P(B) = 0.1$$

$$\begin{aligned} P(\text{saved}) &= 1 - P(\text{lost}) \\ &= 1 - P(H \cap B) \\ &= 1 - P(H)\,P(B) \\ &= 1 - 0.05*0.1 \\ &= 0.995 \end{aligned}$$

Example

- Calculate reliability of the system in Figure if each component is operable with probability 0.92 independently of the other components.



$P(A \cap B) = (0.92)^2 = 0.8464 = P(F)$

$P(D \cup E) = 1 - P\{D \cup E\}'$

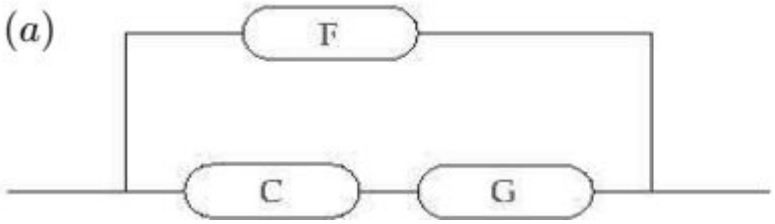
$= 1 - P\{D' \cap E'\}$

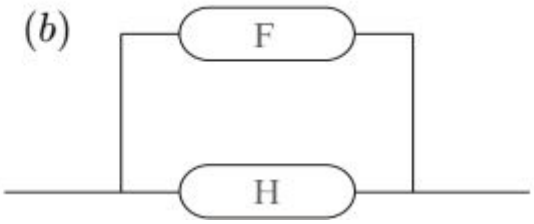
$= 1 - (1 - 0.92)^2 = 0.9936 = P(G)$

$P(C \cap G) = 0.92 \cdot 0.9936 = 0.9141 = P(H)$

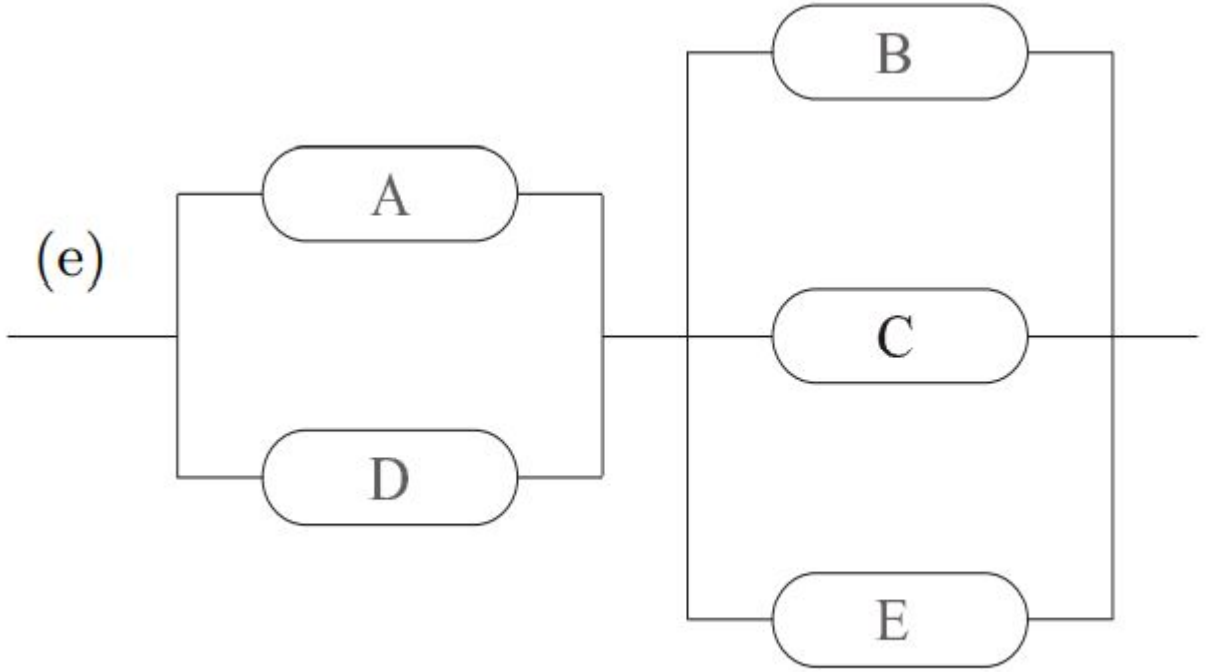
$P(F \cup H) = 1 - (1 - 0.8464)(1 - 0.9141)$

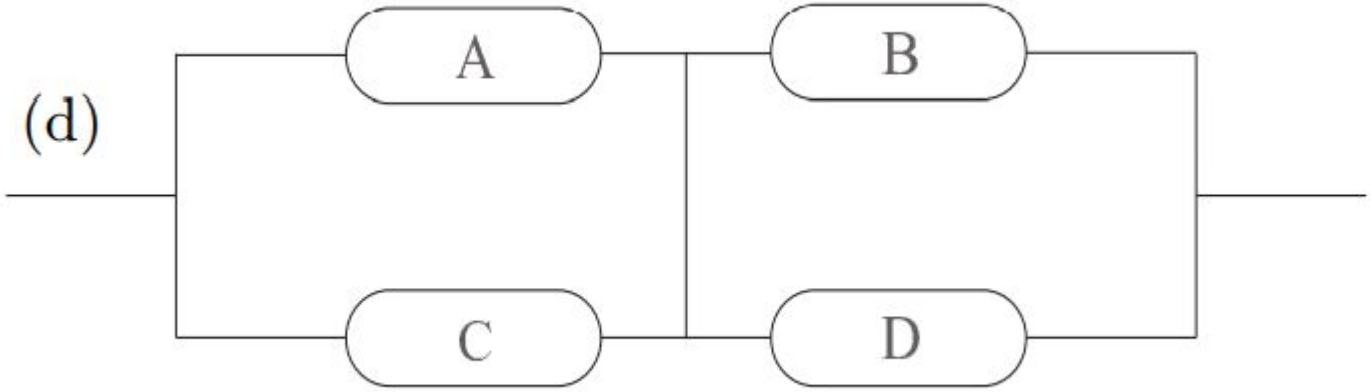
$= 0.9868$





DIY





Rules of Probability

- **Conditional Probability**
  - The conditional probability of *B*, given *A*, is defined by

$$P(B|A) = \frac{P(A \cap B)}{P(A)}$$

The even numbers are twice as likely to occur as the odd numbers.

The probability of B, given A

B = {perfect square} = {1, 4}, P(B) = 1/9 + 2/9 = 1/3

A = {greater than 3} = {4, 5 , 6} ,P(A)=2/9+1/9+2/9=5/9

(A ∩ B)={4} , P(A∩B)= 2/9

No. is a perfect sq, and it is known that no > 3

P(B|A) = (2/9)/(5/9) = 2/5

Rules of Probability

- **Conditional Probability – example**
  - Ninety percent of flights depart on time. Eighty percent of flights arrive on time. Seventy-five percent of flights depart on time and arrive on time.
    - You are meeting a flight that departed on time. What is the probability that it will arrive on time?

    - A = {arriving on time} ,
      D = {departing on time} .
      P (A) = 0.8, P (D) = 0.9, P(A ∩ D) = 0.75.
    - P(A | D) = P (A ∩ D) / P(D) = 0.75 / 0.9 = 0.8333

- One bag contains 4 white balls and 3 black balls, and a second bag contains 3 white balls and 5 black balls. One ball is drawn from the first bag and placed unseen in the second bag. What is the probability that a ball now drawn from the second bag is black?

Rules of Probability

- **General multiplicative rule**
  - $P(A \cap B) = P(A)\, P(B|A) = P(B)\, P(A|B)$

- Two events *A* and *B* are **independent** if and only if
  $P(B|A) = P(B)$ or $P(A|B) = P(A)$
- Three cards are drawn in succession, without replacement, from an ordinary deck of playing cards. Find the probability that the event A1 ∩ A2 ∩ A3 occurs, where A1 is the event that the first card is a red ace, A2 is the event that the second card is a 10 or a jack, and A3 is the event that the third card is greater than 3 but less than 7.

If, in an experiment, the events $A_1, A_2, \ldots, A_k$ can occur, then

$$P(A_1 \cap A_2 \cap \cdots \cap A_k) = P(A_1)P(A_2|A_1)P(A_3|A_1 \cap A_2)\cdots P(A_k|A_1 \cap A_2 \cap \cdots \cap A_{k-1}).$$

If the events $A_1, A_2, \ldots, A_k$ are independent, then

$$P(A_1 \cap A_2 \cap \cdots \cap A_k) = P(A_1)P(A_2)\cdots P(A_k).$$

The property of independence stated in Theorem 2.11 can be extended to deal with more than two events. Consider, for example, the case of three events A, B, and C. It is not sufficient to only have that $P(A \cap B \cap C) = P(A)P(B)P(C)$ as a definition of independence among the three. Suppose $A = B$ and $C = \phi$, the null set. Although $A \cap B \cap C = \phi$, which results in $P(A \cap B \cap C) = 0 = P(A)P(B)P(C)$, events A and B are not independent. Hence, we have the following definition.

A collection of events $\mathcal{A} = \{A_1, \ldots, A_n\}$ are mutually independent if for any subset of $\mathcal{A}$, $A_{i_1}, \ldots, A_{i_k}$, for $k \leq n$, we have

$$P(A_{i_1} \cap \cdots \cap A_{i_k}) = P(A_{i_1})\cdots P(A_{i_k}).$$

Rules of Probability(53B)

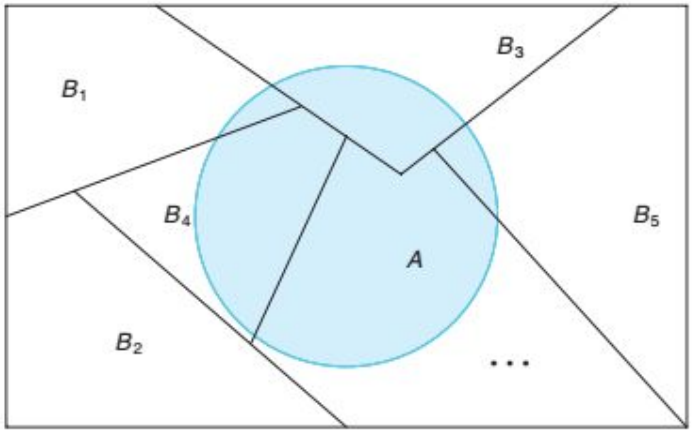
- **Partitions:** A collection of sets $B_1, B_2, \ldots, B_k$ is said to **partition** the sample space if the sets

  (i) are mutually disjoint and **(ii)** have as union the entire sample space.

$$B_i \cap B_j = \varnothing \text{ for any } i \neq j \text{ and } B_1 \cup \ldots \cup B_k = \Omega.$$

- **Total Probability**
  - If the events B1, B2, . . . , Bk constitute a partition of the sample space S such that P (Bi) ≠ 0 for i = 1, 2, . . . , k, then for any event A of S

$$P(A) = \sum_{i=1}^{k} P(A \cap B_i) = \sum_{i=1}^{k} P(A \mid B_i) P(B_i)$$



Partitioning the sample space S.

Random variable

Consider an experiment of tossing 3 fair coins and counting the number of heads.

Let X be the number of heads. Prior to an experiment, its value is not known. All we can say is that X has to be an integer between 0 and 3.

$$\boldsymbol{P}\{X=0\} = \boldsymbol{P}\{\text{three tails}\} = \boldsymbol{P}\{TTT\} = \left(\frac{1}{2}\right)\left(\frac{1}{2}\right)\left(\frac{1}{2}\right) = \frac{1}{8}$$

$$\boldsymbol{P}\{X=1\} = \boldsymbol{P}\{HTT\} + \boldsymbol{P}\{THT\} + \boldsymbol{P}\{TTH\} = \frac{3}{8}$$

$$\boldsymbol{P}\{X=2\} = \boldsymbol{P}\{HHT\} + \boldsymbol{P}\{HTH\} + \boldsymbol{P}\{THH\} = \frac{3}{8}$$

$$\boldsymbol{P}\{X=3\} = \boldsymbol{P}\{HHH\} = \frac{1}{8}$$

x	$\boldsymbol{P}\{X=x\}$
0	1/8
1	3/8
2	3/8
3	1/8
Total	1

Random variable

Two balls are drawn in succession without replacement from an urn containing 4 red balls and 3 black balls. The possible outcomes and the values y of the random variable Y , where Y is the number of red balls, are

Sample Space	y
RR	2
RB	1
BR	1
BB	0

A **random variable** is a function of an outcome,

$$X = f(\omega).$$

A random variable is a function that associates a real number with each element in the sample space.

Random variable

- A random variable is called a discrete random variable if its set of possible outcomes is countable.
- When a random variable can take on values on a continuous scale, it is called a continuous random variable

Probability distributions

A probability distribution provides a **complete picture of all possible outcomes** of a random experiment along with their associated probabilities. It shows how probabilities are distributed across various outcomes, or values, of a random variable.

A probability distribution can either be:

1. Discrete
2. Continuous

Discrete probability distributions

We write f(x) = P(X = x); that is, f(3) = P(X = 3).

The set of ordered pairs (x, f(x)) is a probability function, probability mass function (pmf), or probability distribution of the discrete random variable X if, for each possible outcome x,

1. $f(x) \geq 0$,
2. $\sum_x f(x) = 1$,
3. $P(X = x) = f(x)$.

A shipment of 20 similar laptop computers to a retail outlet contains 3 that are defective. If a school makes a random purchase of 2 of these computers, find the probability distribution for the number of defectives.

$$f(0) = P(X = 0) = \frac{\binom{3}{0}\binom{17}{2}}{\binom{20}{2}} = \frac{68}{95}, \quad f(1) = P(X = 1) = \frac{\binom{3}{1}\binom{17}{1}}{\binom{20}{2}} = \frac{51}{190},$$

$$f(2) = P(X = 2) = \frac{\binom{3}{2}\binom{17}{0}}{\binom{20}{2}} = \frac{3}{190}.$$

Thus, the probability distribution of X is

x	0	1	2
$f(x)$	$\frac{68}{95}$	$\frac{51}{190}$	$\frac{3}{190}$

(Errors in independent modules). A program consists of two modules. The number of errors X_1 in the first module has the pmf $P_1(x)$, and the number of errors X_2 in the second module has the pmf $P_2(x)$, independently of X_1, where

x	$P_1(x)$	$P_2(x)$
0	0.5	0.7
1	0.3	0.2
2	0.1	0.1
3	0.1	0

Find the pmf and cdf of $Y = X_1 + X_2$, the total number of errors.

Solution. We break the problem into steps. First, determine all possible values of Y, then compute the probability of each value. Clearly, the number of errors Y is an integer that can be as low as $0 + 0 = 0$ and as high as $3 + 2 = 5$. Since $P_2(3) = 0$, the second module has at most 2 errors. Next,

$$
\begin{aligned}
P_Y(0) &= P\{Y=0\} &&= \boldsymbol{P}\{X_1 = X_2 = 0\} = P_1(0)P_2(0) \\
&&&= (0.5)(0.7) = 0.35 \\
P_Y(1) &= P\{Y=1\} &&= P_1(0)P_2(1) + P_1(1)P_2(0) \\
&&&= (0.5)(0.2) + (0.3)(0.7) = 0.31 \\
P_Y(2) &= P\{Y=2\} &&= P_1(0)P_2(2) + P_1(1)P_2(1) + P_1(2)P_2(0) \\
&&&= (0.5)(0.1) + (0.3)(0.2) + (0.1)(0.7) = 0.18 \\
P_Y(3) &= P\{Y=3\} &&= P_1(1)P_2(2) + P_1(2)P_2(1) + P_1(3)P_2(0) \\
&&&= (0.3)(0.1) + (0.1)(0.2) + (0.1)(0.7) = 0.12 \\
P_Y(4) &= P\{Y=4\} &&= P_1(2)P_2(2) + P_1(3)P_2(1) \\
&&&= (0.1)(0.1) + (0.1)(0.2) = 0.03 \\
P_Y(5) &= P\{Y=5\} &&= P_1(3)P_2(2) = (0.1)(0.1) = 0.01
\end{aligned}
$$

Now check:

$$\sum_{y=0}^{5} P_Y(y) = 0.35 + 0.31 + 0.18 + 0.12 + 0.03 + 0.01 = 1,$$

thus we *probably* counted all the possibilities and did not miss any (we just wanted to emphasize that simply getting $\sum P(x) = 1$ does not guarantee that we made no mistake in our solution. However, if this equality is not satisfied, we have a mistake for sure).

The cumulative function can be computed as

$$\begin{aligned}
F_Y(0) &= P_Y(0) = 0.35 \\
F_Y(1) &= F_Y(0) + P_Y(1) = 0.35 + 0.31 = 0.66 \\
F_Y(2) &= F_Y(1) + P_Y(2) = 0.66 + 0.18 = 0.84 \\
F_Y(3) &= F_Y(2) + P_Y(3) = 0.84 + 0.12 = 0.96 \\
F_Y(4) &= F_Y(3) + P_Y(4) = 0.96 + 0.03 = 0.99 \\
F_Y(5) &= F_Y(4) + P_Y(5) = 0.99 + 0.01 = 1.00
\end{aligned}$$

Between the values of Y, $F(x)$ is constant. ◊

Cumulative distribution function

There are many problems where we may wish to compute the probability that the observed value of a random variable X will be less than or equal to some real number x.

Writing $F(x) = P(X \leq x)$ for every real number x, we define F(x) to be the cumulative distribution function of the random variable X.

The **cumulative distribution function** $F(x)$ of a discrete random variable X with probability distribution $f(x)$ is

$$F(x) = P(X \leq x) = \sum_{t \leq x} f(t), \quad \text{for } -\infty < x < \infty.$$

A stockroom clerk returns three safety helmets at random to three steel mill employees who had previously checked them. If Smith, Jones, and Brown, in that order, receive one of the three hats, list the sample points for the possible orders of returning the helmets, and find the value m of the random variable M that represents the number of correct matches.

Sample Space	m
SJB	3
SBJ	1
BJS	1
JSB	1
JBS	0
BSJ	0

m	0	1	3
$P(M = m)$	$\frac{1}{3}$	$\frac{1}{2}$	$\frac{1}{6}$

have

$$F(2) = P(M \leq 2) = f(0) + f(1) = \frac{1}{3} + \frac{1}{2} = \frac{5}{6}.$$

The cumulative distribution function of M is

$$F(m) = \begin{cases} 0, & \text{for } m < 0, \\ \frac{1}{3}, & \text{for } 0 \leq m < 1, \\ \frac{5}{6}, & \text{for } 1 \leq m < 3, \\ 1, & \text{for } m \geq 3. \end{cases}$$

Nondecreasing function

Example 3.9: If a car agency sells 50% of its inventory of a certain foreign car equipped with side airbags, find a formula for the probability distribution of the number of cars with side airbags among the next 4 cars sold by the agency.

Find the cumulative distribution function of the random variable X in Example 3.9. Using $F(x)$, verify that $f(2) = 3/8$.

$$f(x) = \frac{1}{16}\binom{4}{x}$$

Direct calculations of the probability distribution of Example 3.9 give $f(0)= 1/16$, $f(1) = 1/4$, $f(2)= 3/8$, $f(3)= 1/4$, and $f(4)= 1/16$. Therefore,

$$F(0) = f(0) = \frac{1}{16},$$
$$F(1) = f(0) + f(1) = \frac{5}{16},$$
$$F(2) = f(0) + f(1) + f(2) = \frac{11}{16},$$
$$F(3) = f(0) + f(1) + f(2) + f(3) = \frac{15}{16},$$
$$F(4) = f(0) + f(1) + f(2) + f(3) + f(4) = 1.$$

$$F(x) = \begin{cases} 0, & \text{for } x < 0, \\ \frac{1}{16}, & \text{for } 0 \le x < 1, \\ \frac{5}{16}, & \text{for } 1 \le x < 2, \\ \frac{11}{16}, & \text{for } 2 \le x < 3, \\ \frac{15}{16}, & \text{for } 3 \le x < 4, \\ 1 & \text{for } x \ge 4. \end{cases}$$

$$f(2) = F(2) - F(1) = \frac{11}{16} - \frac{5}{16} = \frac{3}{8}.$$

Probability histogram

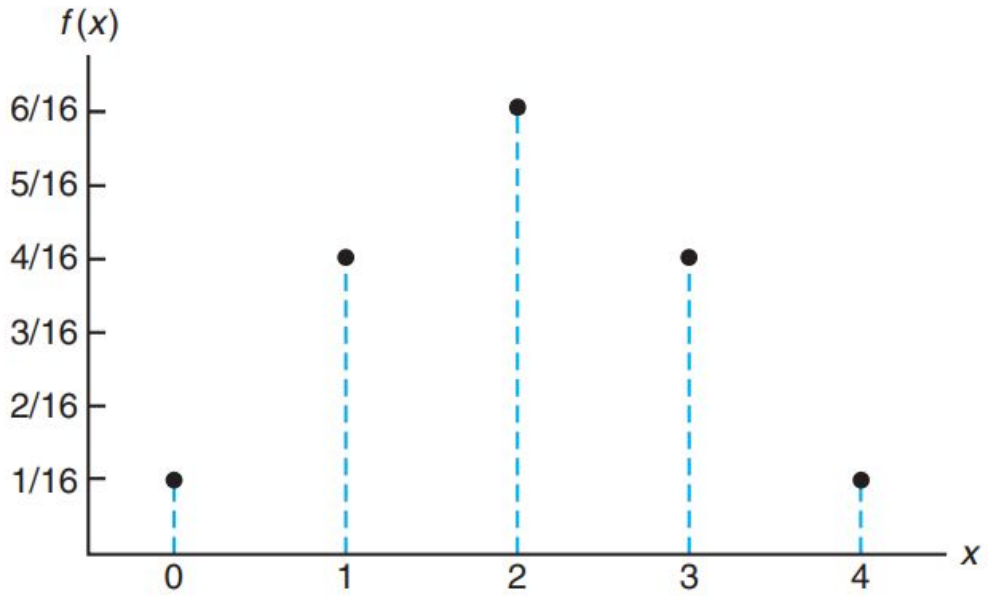


Figure 3.1: Probability mass function plot.

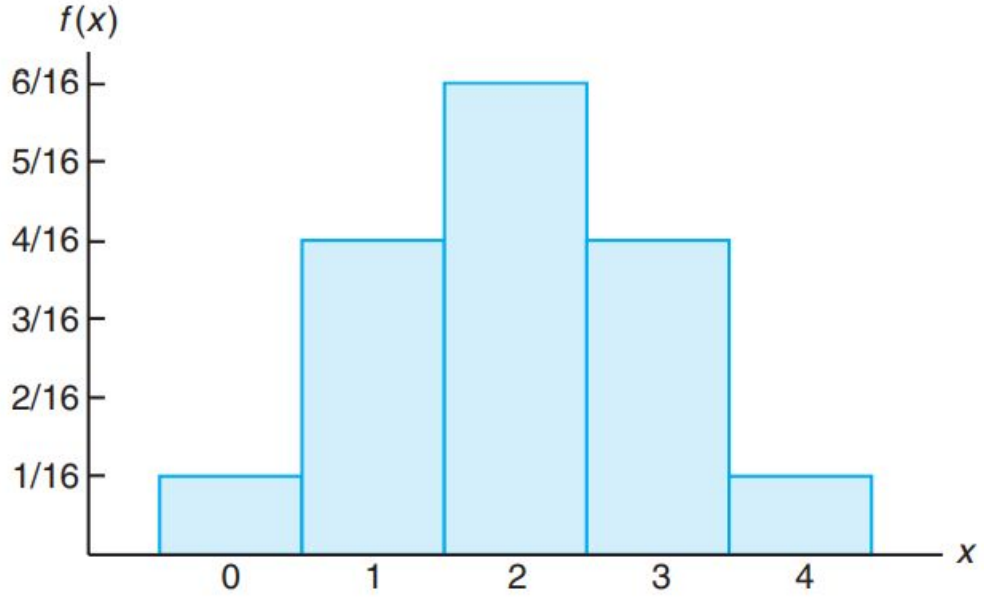


Figure 3.2: Probability histogram.

Continuous probability distributions

Let us discuss a random variable whose values are the heights of all people over 21 years of age. Between any two values, say 163.5 and 164.5 centimeters, or even 163.99 and 164.01 centimeters, there are an infinite number of heights, one of which is 164 centimeters. The probability of selecting a person at random who is exactly 164 centimeters tall and not one of the infinitely large set of heights so close to 164 centimeters that you cannot humanly measure the difference is remote. This is not the case, however, if we talk about the probability of selecting a person who is at least 163 centimeters but not more than 165 centimeters tall.

PDF

We shall concern ourselves with computing probabilities for various intervals of continuous random variables such as $P(a < X < b)$, $P(W \geq c)$, and so forth. Note that when X is continuous,

$$P(a < X \leq b) = P(a < X < b) + P(X = b) = P(a < X < b).$$

That is, it does not matter whether we include an endpoint of the interval or not. This is not true, though, when X is discrete. Although the probability distribution of a continuous random variable cannot be presented in tabular form, it can be stated as a formula. . In dealing with continuous variables, f(x) is usually called the **probability density function**, or simply the density function, of X.

A probability density function is constructed so that the area under its curve bounded by the x axis is equal to 1 when computed over the range of X for which f(x) is defined. In Figure 3.5, the probability that X assumes a value between a and b is equal to the shaded area under the density function between the ordinates at x = a and x = b, and from integral calculus is given by

$$P(a < X < b) = \int_a^b f(x)\,dx.$$

The function $f(x)$ is a **probability density function** (pdf) for the continuous random variable X, defined over the set of real numbers, if

1. $f(x) \geq 0$, for all $x \in R$.
2. $\int_{-\infty}^{\infty} f(x)\,dx = 1$.
3. $P(a < X < b) = \int_a^b f(x)\,dx$.

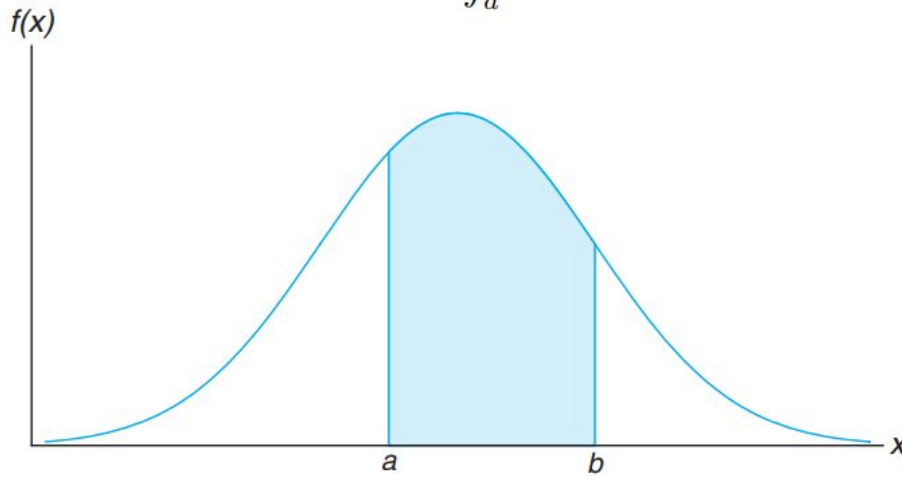


Figure 3.5: $P(a < X < b)$.

Suppose that the error in the reaction temperature, in ◦C, for a controlled laboratory experiment is a continuous random variable X having the probability density function

$$f(x) = \begin{cases} \frac{x^2}{3}, & -1 < x < 2, \\ 0, & \text{elsewhere.} \end{cases}$$

(a) Verify that $f(x)$ is a density function.

(b) Find $P(0 < X \leq 1)$.

(a) Obviously, $f(x) \geq 0$. To verify condition 2 in Definition 3.6, we have

$$\int_{-\infty}^{\infty} f(x)\, dx = \int_{-1}^{2} \frac{x^2}{3} dx = \frac{x^3}{9}|_{-1}^{2} = \frac{8}{9} + \frac{1}{9} = 1.$$

(b) Using formula 3 in Definition 3.6, we obtain

$$P(0 < X \leq 1) = \int_{0}^{1} \frac{x^2}{3} dx = \left. \frac{x^3}{9} \right|_{0}^{1} = \frac{1}{9}.$$

Cumulative distribution function

Definition 3.7:

The **cumulative distribution function** $F(x)$ of a continuous random variable X with density function $f(x)$ is

$$F(x) = P(X \leq x) = \int_{-\infty}^{x} f(t)\, dt, \quad \text{for } -\infty < x < \infty.$$

As an immediate consequence of Definition 3.7, one can write the two results

$$P(a < X < b) = F(b) - F(a) \text{ and } f(x) = \frac{dF(x)}{dx},$$

if the derivative exists.

Example 3.13: The Department of Energy (DOE) puts projects out on bid and generally estimates what a reasonable bid should be. Call the estimate b. The DOE has determined that the density function of the winning (low) bid is

$$f(y) = \begin{cases} \frac{5}{8b}, & \frac{2}{5}b \leq y \leq 2b, \\ 0, & \text{elsewhere.} \end{cases}$$

Find $F(y)$ and use it to determine the probability that the winning bid is less than the DOE's preliminary estimate b.

Joint probability distribution

- If X and Y are two discrete random variables, the probability distribution for their simultaneous occurrence can be represented by a function with values f(x, y) for any pair of values (x, y) within the range of the random variables X and Y . It is customary to refer to this function as the joint probability distribution of X and Y .

$$f(x, y) = P(X = x, Y = y);$$

Joint probability distribution

The values f(x, y) give the probability that outcomes x and y occur at the same time.

For example, if an 18-wheeler is to have its tires serviced and X represents the number of miles these tires have been driven and Y represents the number of tires that need to be replaced, then f(30000, 5) is the probability that the tires are used over 30,000 miles and the truck needs 5 new tires.

The function $f(x, y)$ is a **joint probability distribution** or **probability mass function** of the discrete random variables X and Y if

1. $f(x, y) \geq 0$ for all (x, y),
2. $\sum_x \sum_y f(x, y) = 1$,
3. $P(X = x, Y = y) = f(x, y)$.

For any region A in the xy plane, $P[(X, Y) \in A] = \sum \sum_A f(x, y)$.

Two ballpoint pens are selected at random from a box that contains 3 blue pens, 2 red pens, and 3 green pens. If X is the number of blue pens selected and Y is the number of red pens selected, find

(a) the joint probability function $f(x, y)$,

(b) $P[(X, Y) \in A]$, where A is the region $\{(x, y)|x + y \leq 1\}$.

***Solution*:** The possible pairs of values (x, y) are $(0, 0)$, $(0, 1)$, $(1, 0)$, $(1, 1)$, $(0, 2)$, and $(2, 0)$.

(a) Now, $f(0, 1)$, for example, represents the probability that a red and a green pens are selected. The total number of equally likely ways of selecting any 2 pens from the 8 is $\binom{8}{2} = 28$. The number of ways of selecting 1 red from 2 red pens and 1 green from 3 green pens is $\binom{2}{1}\binom{3}{1} = 6$. Hence, $f(0, 1) = 6/28$

$$f(x, y) = \frac{\binom{3}{x}\binom{2}{y}\binom{3}{2-x-y}}{\binom{8}{2}},$$

Two ballpoint pens are selected at random from a box that contains 3 blue pens, 2 red pens, and 3 green pens. If X is the number of blue pens selected and Y is the number of red pens selected, find

(a) the joint probability function $f(x, y)$,

(b) $P[(X, Y) \in A]$, where A is the region $\{(x, y)|x + y \leq 1\}$.

(b) The probability that (X, Y) fall in the region A is

$$P[(X, Y) \in A] = P(X + Y \leq 1) = f(0, 0) + f(0, 1) + f(1, 0)$$
$$= \frac{3}{28} + \frac{3}{14} + \frac{9}{28} = \frac{9}{14}.$$

Table 3.1: Joint Probability Distribution for Example 3.14

	$f(x, y)$	x = 0	x = 1	x = 2	Row Totals
y	0	$\frac{3}{28}$	$\frac{9}{28}$	$\frac{3}{28}$	$\frac{15}{28}$
	1	$\frac{3}{14}$	$\frac{3}{14}$	0	$\frac{3}{7}$
	2	$\frac{1}{28}$	0	0	$\frac{1}{28}$
Column Totals		$\frac{5}{14}$	$\frac{15}{28}$	$\frac{3}{28}$	1

Example 3.6. A program consists of two modules. The number of errors, X, in the first module and the number of errors, Y, in the second module have the joint distribution, $P(0,0) = P(0,1) = P(1,0) = 0.2$, $P(1,1) = P(1,2) = P(1,3) = 0.1$, $P(0,2) = P(0,3) = 0.05$. Find (a) the marginal distributions of X and Y, (b) the probability of no errors in the first module, and (c) the distribution of the total number of errors in the program. Also, (d) find out if errors in the two modules occur independently.

Solution. It is convenient to organize the joint pmf of X and Y in a table. Adding rowwise and columnwise, we get the marginal pmfs,

$P_{(X,Y)}(x,y)$		y = 0	1	2	3	$P_X(x)$
x	0	0.20	0.20	0.05	0.05	0.50
	1	0.20	0.10	0.10	0.10	0.50
$P_Y(y)$		0.40	0.30	0.15	0.15	1.00

This solves (a).

(b) $P_X(0) = 0.50$.

(c) 1

(d) To decide on the independence of X and Y, check if their joint pmf factors into a product of marginal pmfs. We see that $P_{(X,Y)}(0,0) = 0.2$ indeed equals $P_X(0)P_Y(0) = (0.5)(0.4)$. Keep checking... Next, $P_{(X,Y)}(0,1) = 0.2$ whereas $P_X(0)P_Y(1) = (0.5)(0.3) = 0.15$. There is no need to check further. We found a pair of x and y that violates the formula for independent random variables. Therefore, the numbers of errors in two modules are dependent. ◊

Joint density function

When X and Y are continuous random variables, the **joint density function** $f(x, y)$ is a surface lying above the xy plane, and $P[(X, Y) \in A]$, where A is any region in the xy plane, is equal to the volume of the right cylinder bounded by the base A and the surface.

ition 3.9:

The function $f(x, y)$ is a **joint density function** of the continuous random variables X and Y if

1. $f(x, y) \geq 0$, for all (x, y),
2. $\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y)\, dx\, dy = 1$,
3. $P[(X, Y) \in A] = \int \int_A f(x, y)\, dx\, dy$, for any region A in the xy plane.

Example

A privately owned business operates both a drive-in facility and a walk-in facility. On a randomly selected day, let X and Y, respectively, be the proportions of the time that the drive-in and the walk-in facilities are in use, and suppose that the joint density function of these random variables is

$$f(x,y) = \begin{cases} \frac{2}{5}(2x+3y), & 0 \le x \le 1, 0 \le y \le 1, \\ 0, & \text{elsewhere.} \end{cases}$$

(a) Verify condition 2 of Definition 3.9.

(b) Find $P[(X,Y) \in A]$, where $A = \{(x,y) \mid 0 < x < \frac{1}{2}, \frac{1}{4} < y < \frac{1}{2}\}$.

Solution: (a) The integration of $f(x,y)$ over the whole region is

$$\begin{aligned}\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f(x,y)\,dx\,dy &= \int_0^1\int_0^1 \frac{2}{5}(2x+3y)\,dx\,dy \\ &= \int_0^1 \left(\frac{2x^2}{5}+\frac{6xy}{5}\right)\Big|_{x=0}^{x=1} dy \\ &= \int_0^1 \left(\frac{2}{5}+\frac{6y}{5}\right)dy = \left(\frac{2y}{5}+\frac{3y^2}{5}\right)\Big|_0^1 = \frac{2}{5}+\frac{3}{5}=1.\end{aligned}$$

(b) To calculate the probability, we use

$$\begin{aligned}P[(X,Y)\in A] &= P\left(0<X<\frac{1}{2}, \frac{1}{4}<Y<\frac{1}{2}\right) \\ &= \int_{1/4}^{1/2}\int_0^{1/2} \frac{2}{5}(2x+3y)\,dx\,dy \\ &= \int_{1/4}^{1/2} \left(\frac{2x^2}{5}+\frac{6xy}{5}\right)\Big|_{x=0}^{x=1/2} dy = \int_{1/4}^{1/2}\left(\frac{1}{10}+\frac{3y}{5}\right)dy \\ &= \left(\frac{y}{10}+\frac{3y^2}{10}\right)\Big|_{1/4}^{1/2} \\ &= \frac{1}{10}\left[\left(\frac{1}{2}+\frac{3}{4}\right)-\left(\frac{1}{4}+\frac{3}{16}\right)\right] = \frac{13}{160}.\end{aligned}$$

Given the joint probability distribution $f(x, y)$ of the discrete random variables X and Y, the probability distribution $g(x)$ of X alone is obtained by summing $f(x, y)$ over the values of Y. Similarly, the probability distribution $h(y)$ of Y alone is obtained by summing $f(x, y)$ over the values of X. We define $g(x)$ and $h(y)$ to be the **marginal distributions** of X and Y, respectively. When X and Y are continuous random variables, summations are replaced by integrals. We can now make the following general definition.

Definition 3.10:

The **marginal distributions** of X alone and of Y alone are

$$g(x) = \sum_y f(x, y) \quad \text{and} \quad h(y) = \sum_x f(x, y)$$

for the discrete case, and

$$g(x) = \int_{-\infty}^{\infty} f(x, y)\, dy \quad \text{and} \quad h(y) = \int_{-\infty}^{\infty} f(x, y)\, dx$$

for the continuous case.

Table 3.1: Joint Probability Distribution for Example 3.1

	$f(x,y)$	x = 0	x = 1	x = 2	Row Totals
y	0	$\frac{3}{28}$	$\frac{9}{28}$	$\frac{3}{28}$	$\frac{15}{28}$
	1	$\frac{3}{14}$	$\frac{3}{14}$	0	$\frac{3}{7}$
	2	$\frac{1}{28}$	0	0	$\frac{1}{28}$
Column Totals		$\frac{5}{14}$	$\frac{15}{28}$	$\frac{3}{28}$	1

Show that the column and row totals of Table 3.1 give the marginal distribution of X alone and of Y alone.

***Solution*:** For the random variable X, we see that

$$g(0) = f(0,0) + f(0,1) + f(0,2) = \frac{3}{28} + \frac{3}{14} + \frac{1}{28} = \frac{5}{14},$$

$$g(1) = f(1,0) + f(1,1) + f(1,2) = \frac{9}{28} + \frac{3}{14} + 0 = \frac{15}{28},$$

and

$$g(2) = f(2,0) + f(2,1) + f(2,2) = \frac{3}{28} + 0 + 0 = \frac{3}{28},$$

which are just the column totals of Table 3.1. In a similar manner we could show that the values of $h(y)$ are given by the row totals. In tabular form, these marginal distributions may be written as follows:

x	0	1	2
$g(x)$	$\frac{5}{14}$	$\frac{15}{28}$	$\frac{3}{28}$

y	0	1	2
$h(y)$	$\frac{15}{28}$	$\frac{3}{7}$	$\frac{1}{28}$

Example 3.15: A privately owned business operates both a drive-in facility and a walk-in facility. On a randomly selected day, let X and Y, respectively, be the proportions of the time that the drive-in and the walk-in facilities are in use, and suppose that the joint density function of these random variables is

$$f(x,y) = \begin{cases} \frac{2}{5}(2x+3y), & 0 \le x \le 1, 0 \le y \le 1, \\ 0, & \text{elsewhere.} \end{cases}$$

Find $g(x)$ and $h(y)$ for the joint density function of Example 3.15.

$$g(x) = \int_{-\infty}^{\infty} f(x,y)\,dy = \int_0^1 \frac{2}{5}(2x+3y)\,dy = \left(\frac{4xy}{5} + \frac{6y^2}{10}\right)\Bigg|_{y=0}^{y=1} = \frac{4x+3}{5},$$

for $0 \le x \le 1$, and $g(x) = 0$ elsewhere. Similarly,

$$h(y) = \int_{-\infty}^{\infty} f(x,y)\,dx = \int_0^1 \frac{2}{5}(2x+3y)\,dx = \frac{2(1+3y)}{5},$$

for $0 \le y \le 1$, and $h(y) = 0$ elsewhere.

stochastic process

A **stochastic process** is a random variable that also depends on time. It is therefore a function of two arguments, $X(t, \omega)$, where:

- $t \in \mathcal{T}$ is time, with $\mathcal{T}$ being a set of possible times, usually $[0, \infty)$, $(-\infty, \infty)$, $\{0, 1, 2, \ldots\}$, or $\{\ldots, -2, -1, 0, 1, 2, \ldots\}$;
- $\omega \in \Omega$, as before, is an outcome of an experiment, with Ω being the whole sample space.

Values of $X(t, \omega)$ are called *states*.

Stochastic Process

- Values assumed by X_t are called **states**, set of all possible values of states constitute **state space (*S*)**
- If state space is
  - Discrete – called discrete-state process or **chain**
  - Continuous - called continuous-state process
- If index set T is
  - Discrete – called discrete-time process or **sequence**
  - Continuous - called continuous-time process

Markov Chain

- Markov Property: a stochastic process is said to have ~ if probability distribution of future state depends only on present state and not on how the process arrived in that state.
- Formally-The state of the system at time t+1 depends only on the state of the system at time t

$$P[X_{t+1} = x_{t+1} \mid X_t = x_t, X_{t-1} = x_{t-1}, \ldots X_0 = x_t] = P[X_{t+1} = x_{t+1} \mid X_t = x_t]$$

- A stochastic process $\{X_t\}$ having Markov property is called **Markov Process**
  - **Markov chain** - If the state space of a Markov process is discrete
  - If index set is also discrete – **Discrete-time Markov chain**

Stationary Markov Chain

- **Stationary Assumption:** For all states *i, j* and for all *t*, $P(X_{t+1} = j \mid X_t = i)$ is independent of time (t)

$$P(X_{t+1} = j \mid X_t = i) = P(X_1 = j \mid X_0 = i) = p_{ij}$$

$$; \quad i,j = 0,1, \ldots, s; \quad t = 0,1, \ldots,T$$

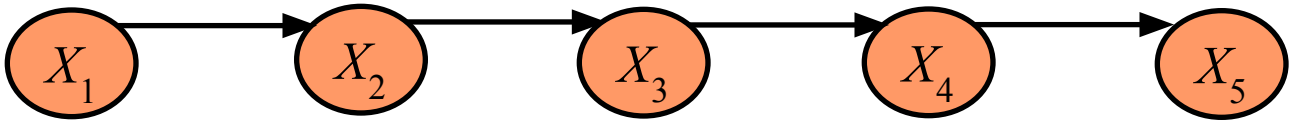
- This means that if system is in state i, the probability that the system will transition to state j is pij no matter what the value of t is
- p_{ij} = probability that the system will be in state j at time t+1 given that it is in state *i* at time t
  - Called one step **transition probability**
- Stationary Markov Chain – having stationary transition probabilities

Markov Chain

- We generally represent transition probabilities of Markov Chain by a $s \times s$ **Transition Probability Matrix P**

$$P = \begin{bmatrix} p_{11} & p_{12} & \cdots & p_{1s} \\ p_{21} & p_{22} & \cdots & p_{2s} \\ \vdots & \vdots & & \vdots \\ p_{s1} & p_{s2} & \cdots & p_{ss} \end{bmatrix} \qquad p_{ij} \geq 0 \quad \text{and} \quad \sum_{j=1}^{s} p_{ij} = 1$$

- It can also be represented by stochastic Finite state Machine



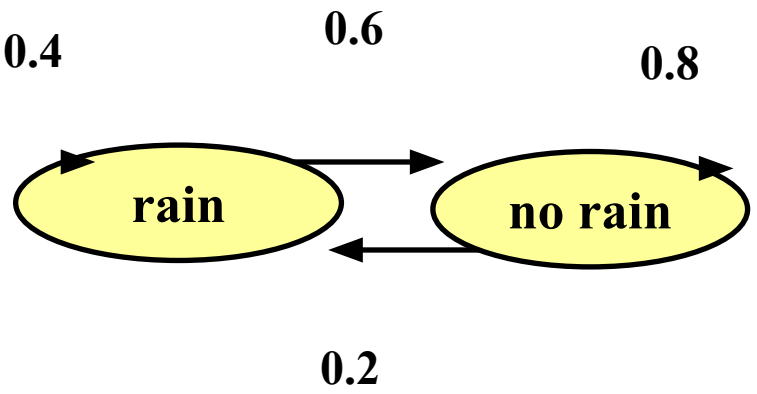
We call the vector q [q1 q2 . . . qs] the initial probability distribution for the Markov chain.

Markov Chain
Simple Example

Weather:

- raining today → rain tomorrow
  60% no rain tomorrow
- not raining today → 20% rain tomorrow
  80% no rain tomorrow

Stochastic FSM:



Transition Prob. Matrix:

State 1: Rain
State 2: No Rain

$$P = \begin{pmatrix} 0.4 & 0.6 \\ 0.2 & 0.8 \end{pmatrix}$$

Transforming a process to a Markov chain

- Whether or not it rains today depends on previous weather conditions through last two days
  - If it rained for past two days, it will rain tomorrow with prob. 0.7
  - If it rained today but not yesterday, it will rain tomorrow with prob. 0.5
  - If it rained yesterday but not today, it will rain tomorrow with prob. 0.4
  - If it has not rained for past two days, it will rain tomorrow with prob. 0.2
- Let the state at time n – depend only on a single day
  - Not Markov chain
- Convert – n saying that it depend on both day

Transforming a process to a Markov chain

- State 0 - If it rained today and yesterday (RR)
- State 1 - If it rained today but not yesterday (NR)
- State 2 - If it rained yesterday but not today (RN)
- State 3 - If it did not rained either today or yesterday (NN)

Yesterday, Today	Today, Tomorrow RR	NR	RN	NN
RR	0.7	0	0.3	0
NR	0.5	0	0.5	0
RN	0	0.4	0	0.6
NN	0	0.2	0	0.8

$$P = \begin{bmatrix} 0.7 & 0 & 0.3 & 0 \\ 0.5 & 0 & 0.5 & 0 \\ 0 & 0.4 & 0 & 0.6 \\ 0 & 0.2 & 0 & 0.8 \end{bmatrix}$$

Markov Chain

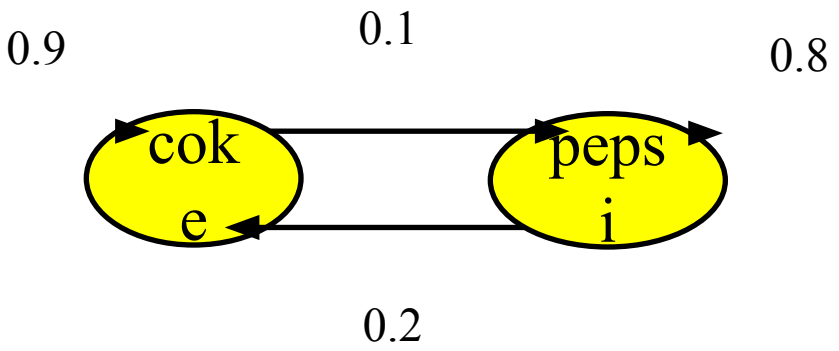
Coke vs. Pepsi Example

- Given that a person's last cola purchase was Coke, there is a 90% chance that his next cola purchase will also be Coke.
- If a person's last cola purchase was Pepsi, there is an 80% chance that his next cola purchase will also be Pepsi.

transition matrix:

$$P = \begin{bmatrix} 0.9 & 0.1 \\ 0.2 & 0.8 \end{bmatrix}$$

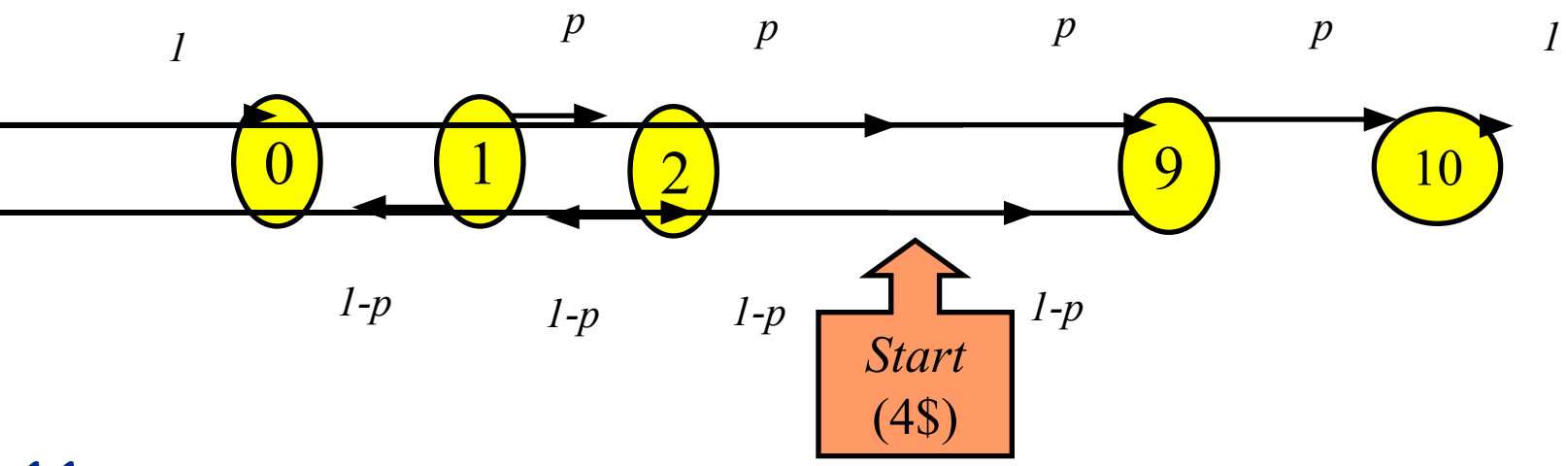
State 1: Coke
State 2: Pepsi



Markov Chain
Gambler's Example

- Gambler starts with $4
- At each play we have one of the following:
  - Gambler wins $1 with probability p
  - Gambler looses $1 with probability $1-p$
- Game ends when gambler goes broke, or gains a fortune of $10

  (Both 0 and 10 are absorbing states)



Characteristics of a Markov chain

- What do we need to know to describe a Markov chain?
- Next state depends on previous state only, therefore, it is sufficient to know
  - the distribution of its initial state X_0
    - $P_0(x) = P\{X(0) = x\}$ for $x \in \{1, 2, \ldots, n\}$
    - initial distribution P_0 – pmf of X_0
  - the mechanism of transitions from one state to another.
    - one-step transition probabilities p_{ij}.

Characteristics of a Markov chain

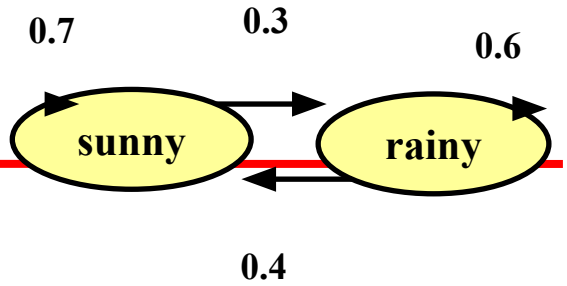
- Based on this data, we would like to compute:
  - n-step transition probabilities $\boldsymbol{p_{ij}(n)}$;
  - Q_n the distribution of states at time n, which is our forecast for X_n;
  - The limit of $\boldsymbol{p_{ij}(n)}$; and Q_n as $n \to \infty$, which is our long-term forecast.

NOTATION

$$p_{ij} = P\{X(t+1) = j \mid X(t) = i\},$$
transition probability

$$p_{ij}^{(h)} = P\{X(t+h) = j \mid X(t) = i\},$$
h-step transition probability

n-step Transition Probabilities



- We may be interested in
  - It rains on Monday. Make forecasts for Wednesday, and Thursday. (For weather forecast example with above FSM)
- Mathematically
  - $p_{21}(2) = P\{$ Wednesday is sunny | Monday is rainy $\}$
- More generally
  - If a Markov chain in state *i* at time m, what is the probability that n periods later the Markov chain will be in state *j*
  - ie. $P(X_{m+n} = j \mid X_m = i) = ?$

n-step Transition Probabilities

- Since we are dealing with stationary Markov chain, we can write
  - $P(X_{m+n} = j \mid X_m = i) = P(X_n = j \mid X_0 = i) = p_{ij}(n)$
    - $p_{ij}(n)$ = n-step probability of transition from state *i* to state *j*

- *If*

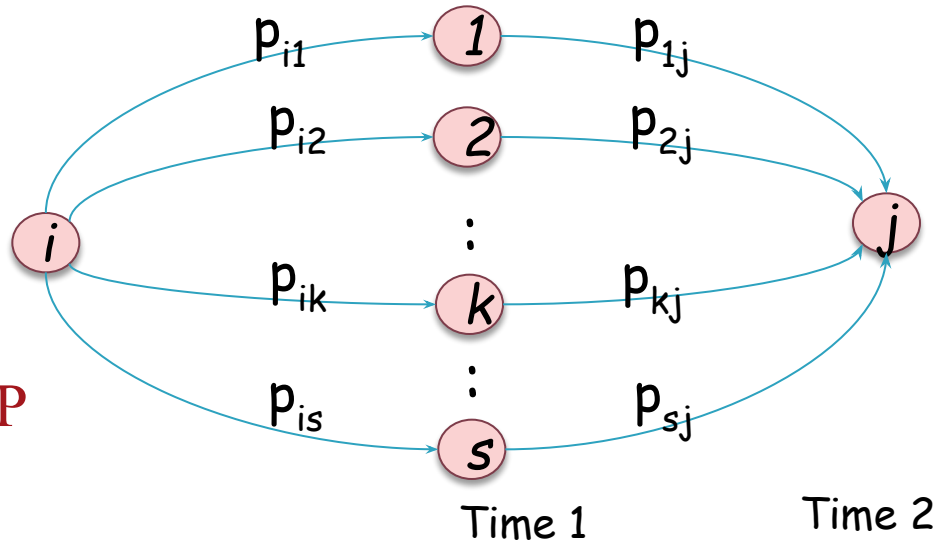
$$P = \begin{bmatrix} p_{11} & p_{12} & \cdots & p_{1s} \\ p_{21} & p_{22} & \cdots & p_{2s} \\ \vdots & \vdots & & \vdots \\ p_{s1} & p_{s2} & \cdots & p_{ss} \end{bmatrix} \qquad p_{ij} \geq 0 \quad \text{and} \quad \sum_{j=1}^{s} p_{ij} = 1$$

- Cleary $p_{ij}(1) = p_{ij}$
- Matrix $\mathbf{P^n}$ represents n-step transition probabilities from any state *i* to state *j*
  - How will you find P^n ?

n-step Transition Probabilities

- $p_{ij}(2)$= probability that the system will be in state j two periods from now, considering it is now in state i
- To compute $p_{ij}(2)$, we must go from state i to some state k, then form state k to state j

$$p_{ij}(2) = \sum_{k=1}^{s} p_{ik}\, p_{kj} \quad \text{for all states } i, j$$



- Clearly right side of the eq^{n} is the scalar product of ith row of matrix P with j^{th} column of matrix P
- Hence, $p_{ij}(2)$ is the ij^{th} element of matrix P.P = $\mathbf{P^2}$
- i.e. Matrix $\mathbf{P^2}$ represents 2-step transition probabilities for all states i, j

Markov chain

Weather example **(cont)**

$p_{21}(2)$ = P { Wednesday is sunny | Monday is rainy }

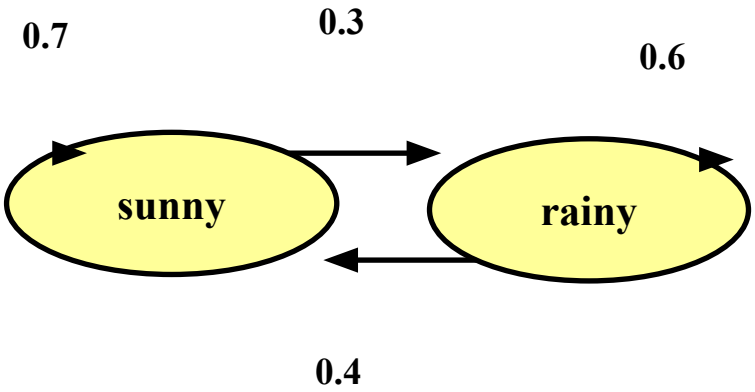
$= p_{21}p_{11} + p_{22}p_{21}$

$= (0.4)(0.7) + (0.6)(0.4)$

$= 0.52.$

state 1 = "sunny"

state 2 = "rainy.



Markov chain

Coke vs. Pepsi Example (cont)

Given that a person is currently a Pepsi purchaser, what is the probability that he will purchase Coke two purchases from now?

Pr[Pepsi → ? → Coke] =

Pr[Pepsi → Coke → Coke] + Pr[Pepsi → Pepsi → Coke] =

$0.2 * 0.9 + 0.8 * 0.2 = 0.34$

$$P = \begin{bmatrix} 0.9 & 0.1 \\ 0.2 & 0.8 \end{bmatrix} \begin{bmatrix} 0.9 & 0.1 \\ 0.2 & 0.8 \end{bmatrix} = \begin{bmatrix} 0.83 & 0.17 \\ 0.34 & 0.66 \end{bmatrix}$$

Pepsi → ? ? → Coke

Markov chain

Coke vs. Pepsi Example (cont)

- Given that a person is currently a Coke purchaser, what is the probability that he will purchase Pepsi **three** purchases from now?

$$P^3 = \begin{bmatrix} 0.9 & 0.1 \\ 0.2 & 0.8 \end{bmatrix} \begin{bmatrix} 0.83 & 0.17 \\ 0.34 & 0.66 \end{bmatrix} = \begin{bmatrix} 0.781 & 0.219 \\ 0.438 & 0.562 \end{bmatrix}$$

Queuing System (elements)

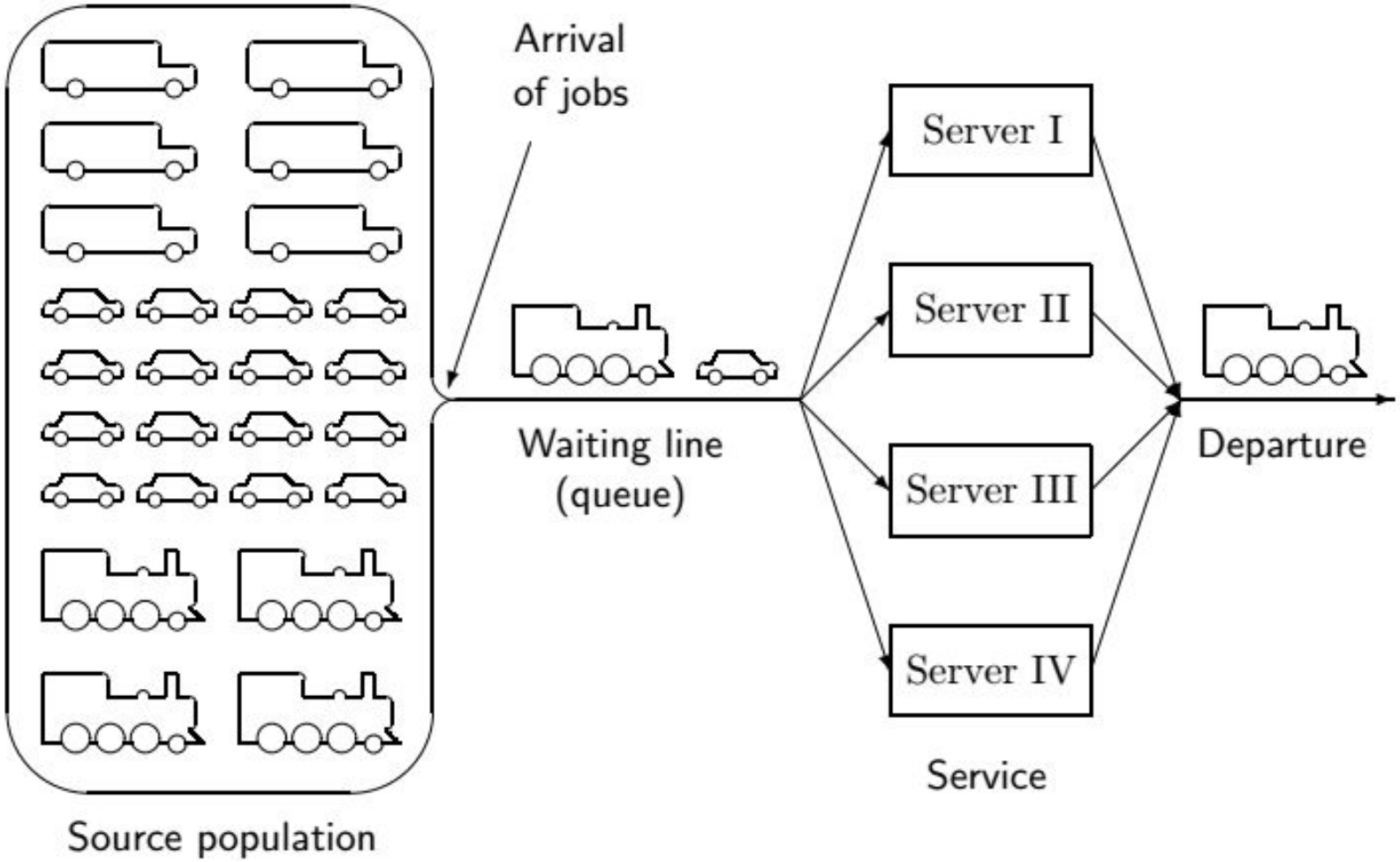
- Waiting for service is part of our daily life. We queue up for
  - Bus, check out counter etc.
- Not only for human, jobs wait to be processed on a machine
- Study of queue quantifies the phenomenon of waiting lines using
  - Average queue length, average waiting time, average facility utilization
  - Thereby reduce the adverse impact of waiting to tolerable levels

Queuing System

- A queuing system is a facility consisting of one or several servers designed to perform certain tasks or process certain jobs and a queue of jobs waiting to be processed.
- Examples of queuing systems are:
  - a shared computer executing tasks sent by its users;
  - an internet service provider whose customers connect to the internet, browse, and disconnect;
  - a printer processing jobs sent to it from different computers;
  - a medical office serving patients; and so on.

Queuing System

- Main components



Queuing System - Input process

- Usually called the **arrival process**.
- Arrivals are called **customers**.
- We assume that no more than one arrival can occur at a given instant.
- If more than one arrival can occur at a given instant, we say that **bulk arrivals** are allowed.
- Models in which arrivals are drawn from a small population are called **finite source models**.
- If a customer arrives but fails to enter the system, we say that the customer has **balked**.

Queuing System - Output process

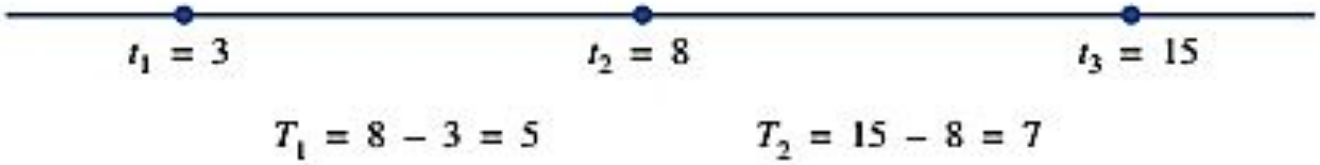
- To describe the output process of a queuing system, we usually specify a probability distribution – the **service time distribution** – which governs a customer's service time.
- Two arrangements of servers:
  - Servers are in **parallel** if all server provide the same type of service and a customer need only pass through one server to complete service.
  - Servers are in **series** if a customer must pass through several servers before completing service.

Queuing System - Queue discipline

- Describes the method used to determine the order in which customers are served.
- The most common - **FCFS discipline** (first come, first served), served in order of their arrival.
- **LCFS discipline** (last come, first served), the most recent arrivals are the first to enter service.
- Customer is randomly chosen from waiting line - **SIRO discipline** (service in random order).
- **Priority queuing disciplines -** classifies each arrival into one of several categories.
  - Each category is then given a priority level, and within each priority level served on an FCFS basis.

Modeling Arrival and Service Processes

- We define t_i to be the time at which the ith customer arrives.
- we define $T_i = t_{i+1} - t_i$ to be the ith interarrival time.
- In modeling the arrival process we assume that the T_i's are independent, continuous random variables described by **A** having a density function $a(t)$.
- The assumption that each interarrival time is governed by the same random variable implies that the distribution of arrivals is independent of the time of day or the day of the week.
- This is the assumption of stationary interarrival times.
- Stationary-(whose distribution characteristics do not change over time)



The Kendall-Lee Notation for Queuing Systems

- Standard notation used to describe many queuing systems.
- The notation is used to describe a queuing system in which all arrivals wait in a single line until one of *s* identical parallel servers id free. Then the first customer in line enters service, and so on.
- To describe such a queuing system, Kendall devised the following notation.
- Each queuing system is described by six characters:

  1/2/3/4/5/6

The Kendall-Lee Notation for Queuing Systems

- The first characteristic specifies the nature of the arrival process. The following standard abbreviations are used:

  M = Interarrival times are independent, identically distributed (iid) having an exponential distribution.

  D = Interarrival times are iid and deterministic

  E_k = Interarrival times are iid Erlangs with shape parameter k.

  GI = Interarrival times are iid and governed by some general distribution

The Kendall-Lee Notation for Queuing Systems

- The second characteristic specifies the nature of the service times:

  M = Service times are iid and exponentially distributed

  D = Service times are iid and deterministic

  E_k = Service times are iid Erlangs with shape parameter k.

  G = Service times are iid and governed by some general distribution

The Kendall-Lee Notation for Queuing Systems

- The third characteristic is the number of parallel servers.
- The fourth characteristic describes the queue discipline:
  - FCFS = First come, first served
  - LCFS = Last come, first served
  - SIRO = Service in random order
- The fifth characteristic specifies the maximum allowable number of customers in the system.
- The sixth characteristic gives the size of the population from which customers are drawn.

The Kendall-Lee Notation for Queuing Systems

- In many important models 4/5/6 is $GD/\infty/\infty$. If this is the case, then 4/5/6 is often omitted.
- $M/E_2/8/\text{FCFS}/10/\infty$ might represent a health clinic with 8 doctors, exponential interarrival times, two-phase Erlang service times, an FCFS queue discipline, and a total capacity of 10 patients.

Birth-Death Processes

- We will use it to answer questions about several different types of queuing systems.
- Number of people present in any queuing system at time *t* to be the **state** of the queuing systems at time *t*.
- $P_{ij}(t)$ which is defined as the probability that j people will be present in the queuing system at time t, given that at time 0, i people are present.
- We call π_j the **steady state**, or equilibrium probability, of state *j*.
- The behavior of $P_{ij}(t)$ before the steady state is reached is called the **transient behavior** of the queuing system.
- A **birth-death process** is a continuous-time stochastic process for which the system's state at any time is a nonnegative integer.

Laws of Motion for Birth-Death

- Law 1
  - With probability $\lambda_j \Delta t + o(\Delta t)$, a birth occurs between time t and time $t+\Delta t$.
  - A birth increases the system state by 1, to $j+1$.
  - The variable λ_j is called the **birth rate** in state j.
  - In most queuing systems, a birth is simply an arrival.
- Law 2
  - With probability $\mu_j \Delta t + o(\Delta t)$, a death occurs between time t and time $t + \Delta t$. A death decreases the system state by 1, to j-1. The variable μ_j is the death rate in state j. In most queuing systems, a death is a service completion.
  - Note that $\mu_0 = 0$ must hold, or a negative state could occur.
- Law 3
  - Births and deaths are independent of each other.